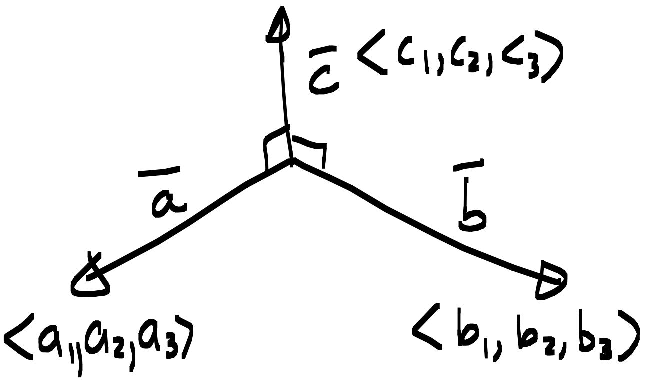
Unit 6.4 The cross product p814 - 821



$$\left.\begin{aligned} \bar{a}\cdot\bar{c} &= 0 \\ \bar{b}\cdot\bar{c} &= 0 \end{aligned}\right\}$$ and

$$\left[\begin{aligned} a_1 c_1 + a_2 c_2 + a_3 c_3 &= 0 \quad \boxed{1} \\ b_1 c_1 + b_2 c_2 + b_3 c_3 &= 0 \quad \boxed{2} \end{aligned}\right. \qquad (c_1, c_2, c_3)$$

Check the book p 808

$$\bar{c} = \langle c_1, c_2, c_3 \rangle = \langle \underbrace{a_2 b_3 - a_3 b_2, a_3 b_1 - a_1 b_3, a_1 b_2 - a_2 b_1}_{\text{Yech!!}} \rangle \quad (*)$$

Definition: Let a, b, c and $d \in \mathbb{R}$.

A determinant of order 2 is

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc \qquad (2\times 2)$$

1st

$$\det\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

Example:

(1) $$\begin{vmatrix} -1 & 2 \\ 3 & 4 \end{vmatrix} = -10$$

(2) $\begin{vmatrix} 1 & 1 & 1 \\ 3 & 1 & 2 \\ 0 & 1 & 1 \end{vmatrix} = 1\begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix} - 3\begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} + 0\begin{vmatrix} 1 & 1 \\ 1 & 2 \end{vmatrix}$

$$= 1(1-2) - 3(1-1) + 0(2-1) = -1$$

Definition:

If $\bar{a} = \langle a_1, a_2, a_3 \rangle$ and $\bar{b} = \langle b_1, b_2, b_3 \rangle$ then the crossproduct of $\bar{a}$ and $\bar{b}$ is

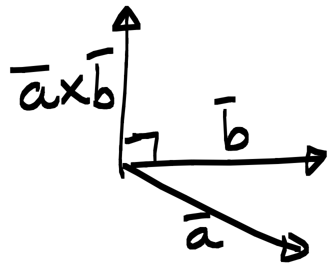
$$\bar{a} \times \bar{b} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

$$= \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} \bar{i} - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} \bar{j} + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \bar{k}$$

$$= (a_2 b_3 - b_2 a_3)\, i - \ldots\ldots \text{ etc.}$$

Compare to (*)

$\bar{a} \times \bar{b}$ is a vector orthogonal to $\bar{a}$ and $\bar{b}$



$$(\bar{a} \times \bar{b}) \cdot \bar{b} = 0$$

$$(\bar{a} \times \bar{b}) \cdot \bar{a} = 0$$

Example:

Determine $\bar{u} \times \bar{v}$ if $\bar{u} = \langle -3, 4, -2 \rangle$ and $\bar{v} = \langle 2, -1, 3 \rangle$

$$\underline{\underline{\bar{u} \times \bar{v}}} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ -3 & 4 & -2 \\ 2 & -1 & 3 \end{vmatrix} = i \begin{vmatrix} 4 & -2 \\ -1 & 3 \end{vmatrix} - j \begin{vmatrix} -3 & -2 \\ 2 & 3 \end{vmatrix} + k \begin{vmatrix} -3 & 4 \\ 2 & -1 \end{vmatrix}$$

$$= \ldots\ldots$$

vector

$$= \langle 10, 5, -5 \rangle = 10\bar{i} + 5\bar{j} - 5\bar{k}$$

$$(\bar{u} \times \bar{v}) \cdot \bar{u} = \langle 10, 5, -5 \rangle \cdot \langle -3, 4, -2 \rangle$$

$$= 0$$

orthogonal.

number

Summary $\bar{a}, \bar{b} \in \mathbb{R}^3$, $\bar{a} \neq 0$, $\bar{b} \neq 0$

1. $\bar{a} \times \bar{b}$ is orthogonal on both $\bar{a}$ and $\bar{b}$
2. $\bar{a} \parallel \bar{b} \Rightarrow \bar{a} \times \bar{b} = \bar{0}$
3. $|\bar{a} \times \bar{b}|$ is the area of the paralellogram through $\bar{a}$ and $\bar{b}$

Examples

1. Let $\bar{a} = \langle 1, 2, 0 \rangle$ and $\bar{b} = \langle 3, -1, 2 \rangle$

   $\bar{a} \times \bar{b} =$

2. $\bar{a} = \langle 2, 3, -1 \rangle$ and $\bar{b} = \langle -4, -6, 2 \rangle$

   $\bar{a} \times \bar{b} =$

3. Determine a vector with length 2 orthogonal to $\bar{a} = \langle 1, -1, 1 \rangle$ and $\bar{b} = \langle 3, 2, -1 \rangle$

4. Determine the area of the Δ with vertices $A(2, -3, 4)$, $B(1, 2, 3)$ and $C(0, -2, -1)$

Properties of the crossproduct p 819

$\bar{a}, \bar{b}, \bar{c} \in \mathbb{R}^3$ and $k \in \mathbb{R}$

1. $\bar{a} \times \bar{b} = -\bar{b} \times \bar{a}$
2. $(k\bar{a}) \times \bar{b} = k(\bar{a} \times \bar{b})$
3. $\bar{a} \times (\bar{b} + \bar{c}) = \bar{a} \times \bar{b} + \bar{a} \times \bar{c}$
4. $(\bar{a} + \bar{b}) \times \bar{c} = \bar{a} \times \bar{c} + \bar{b} \times \bar{c}$